

## Pascal's triangle



1 The 4th row of Pascal's triangle consists of the numbers 1, 4, 6, 4, 1.

- a Write down the numbers in the 5th row of Pascal's triangle.
- b Find the sum of the entries in the 5th row of Pascal's triangle.
- c Rewrite each element of the 5th row of Pascal's triangle as an  ${}_nC_r$  coefficient.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1, 8, \mathbb{I}, 56, 70, \mathbb{I}, 28, \mathbb{I}, 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the distribution of  $(x + y)^7$ ?

4 State the number of terms in the following distributions:

a  $(4x + y)^6$

b  $(m + y)^8$

5 Consider the distribution of  $(y + k)^4$ .

- a How many terms will there be in the distribution?
- b Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following distributions:

a  $(1 + x)^4 = \mathbb{I} + \mathbb{I}x + \mathbb{I}x^2 + \mathbb{I}x^3 + \mathbb{I}x^4$

b  $(x + m)^4 = \mathbb{I}x^4m^0 + \mathbb{I}x^3m^1 + \mathbb{I}x^2m^2 + \mathbb{I}x^1m^3 + \mathbb{I}x^0m^4$

c  $(5 + b)^5 = \mathbb{I} \times 5^5b^0 + \mathbb{I} \times 5^4b^1 + \mathbb{I} \times 5^3b^2 + \mathbb{I} \times 5^2b^3 + \mathbb{I} \times 5^1b^4 + \mathbb{I} \times 5^0b^5$

d  $(ka - b)^6 = \mathbb{I}(ka)^4(-b)^0 + \mathbb{I}(ka)^3(-b)^1 + \mathbb{I}(ka)^2(-b)^2 + \mathbb{I}(ka)^1(-b)^3$   
 $+ \mathbb{I}(ka)^0(-b)^4 + \mathbb{I}(ka)^0(-b)^5 + \mathbb{I}(ka)^0(-b)^6$

7 Distribute the following using Pascal's triangle:

a  $(p + q)^5$

b  $(c - d)^8$

c  $(y + 3)^4$

d  $(u + 4v)^6$

e  $\left(1 + \frac{4}{y}\right)^3$

8 Evaluate the following:



a  $\binom{6}{5}$

b  $\binom{14}{1}$

c  $\binom{10}{9}$

d  $\binom{8}{0}$

## Binomial theorem

9 Consider the distribution of  $(a - 4)^5$ . Will the 2nd term be positive or negative?

10 Complete the following distribution:

$$(4p + 3q)^3 = {}_3C_0 (4p)^0 (3q)^3 + {}_3C_1 (4p)^1 (3q)^2 + {}_3C_2 (4p)^2 (3q)^1 + {}_3C_3 (4p)^3 (3q)^0$$

11 Consider the binomial  $(4x + 3y)^4$ .

a State the first term in the distribution.

b State the last term in the distribution.

12 Which two terms in the distribution of  $(u + v)^{11}$  have a coefficient of  ${}_{11}C_9$ ?

13 Find the specified term for the following distributions:

a Fifth term of  $(b + 2)^7$ .

b Fifth term of  $(c + d)^9$ .

c Fifth term of  $(3u - v)^6$ .

d Tenth term of  $\left(u + \frac{1}{2}\right)^{12}$ .

e Seventh term of  $\left(\frac{3y}{2} - \frac{2}{3y}\right)^{10}$ .

f Eleventh term of  $(2x + y^2)^{13}$ .

g Sixteenth term of  $(x - y^5)^{19}$ .

h Middle term of  $(4x^2 + 3y^3)^8$ .

i Middle term of  $(-3x^{-3} + 2y^{-2})^4$ .

j Constant term of  $\left(2y - \frac{1}{y^3}\right)^8$ .

14 Use the binomial theorem to distribute the following expressions:

a  $(1 + x)^4$

b  $(2 + b)^5$

c  $(p + q)^5$

d  $(c - d)^8$

e  $(u + 4v)^6$

f  $(y + 3)^4$

g  $(ka - b)^6$

h  $(y - 5)^3$

i  $\left(y - \frac{1}{3}\right)^3$

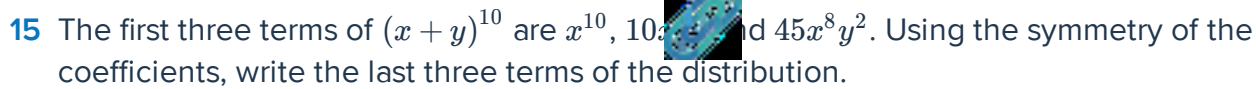
j  $\left(2 + \frac{y}{2}\right)^3$

k  $(3y + 2r)^3$

l  $\left(a - \frac{1}{a}\right)^3$

m  $(u^2 + 3v^2)^3$

n  $\left(1 + \frac{4}{y}\right)^3$



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- 26** Complete the following distribution by determining the missing power and binomial coefficient of the  $(k + 1)$ th term:

$$(3 - x)^8 = {}_8C_0 \times 3^8 (-x)^0 + {}_8C_1 \times 3^7 (-x)^1 + \dots + \mathbb{I} \times 3^{8-k} (-x)^{\mathbb{I}} + \dots + {}_8C_8 \times 3^0 (-x)^8$$

- 27** Consider the binomial series:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

- a** Use the first four terms to approximate  $(1.03)^{-2}$  to the nearest thousandth.
- b** Use the first four terms to approximate  $(1.05)^{\frac{3}{4}}$  to the nearest thousandth.
- c** Use the first five terms to approximate  $\frac{1}{(1.06)^2}$  to the nearest thousandth.